

Supplementary S4. Reference-record, aggregated-weight, and edge-count contract

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Unit-aware audit of frozen counts and weights

stage / class	count	unit	measure
before admission	3,748,078	RECORD	-
admitted	3,747,958	RECORD	-
out of seed	120	RECORD	-
missing repo ID	0	RECORD	-
invalid repo ID	0	RECORD	-
target mappable	1,586,047	RECORD	-
target non-project	1,686,729	RECORD	-
target unresolved	475,182	RECORD	-
target ambiguous	0	RECORD	-
conflict excluded	0	RECORD	-
quotient eligible	1,586,047	RECORD	-
self-loop evidence	1,447,073	REFERENCE_RECORD	aggregated edge weight
cross-project evidence	138,974	REFERENCE_RECORD	aggregated edge weight
self-loop edges	289	EDGE_COUNT	self-loop edge count
cross-project directed edge	9,595	EDGE_COUNT	cross-project edge count
Cross Project Reference	138,974	REFERENCE_RECORD	edge-class weight
Not Quotient Eligible Refer	2,161,911	REFERENCE_RECORD	edge-class weight
Self Internal Project Refer	1,447,073	REFERENCE_RECORD	edge-class weight

RECORD counts, REFERENCE_RECORD/aggregated RefQ weights, and EDGE_COUNT values are separate contracts; no common quantitative axis is implied.